

Purpose

This document describes cohorting procedure and policy for patients with Multi-Drug Organisms (MDRO) colonization or infection. In addition, it applies to other patients e.g. cases with same respiratory disease such as Influenza, ext.

Scope:

The cohort option only applies to patients for contact or droplet isolations. It does not apply to patients with suspected/confirmed airborne infections e.g. open TB cases and chickenpox.

Roles and responsibilities:

- It is the responsibility of all Healthcare Workers to comply with this policy and to follow infection control precautions.
- The infection prevention and control will conduct surveillance for Multi Drug Resistant Organisms (MDROs) such as MRSA, MDRA, CRE, VRE, Candida auris and PAER.
- The Microbiology laboratory will generate all MDROs positive patients list which must be reviewed by the ICP on the working days.
- The ICP must notify the status, clusters or cross transmission of the MDROs to the concerned departments.
- Daily MDRO colonized patients List should be generated by each ward (in medical, surgical and NOC units) in morning shift. The list is communicated to ICP (covering cohort ward), Cohort ward nurse and bed manager team to createlist of patients for cohort.
- Bed managers to liaise with the concerned patients' team to assess their clinical status if fit to be transferred to cohort area. Documentation of fitness of the patient to be cohorted should be done by the caring team in the patient's progress notes.
- It is the responsibility of each unit to initiate cohort upon notification by the ICP in liaison with bed manager team.
- During weekends and holidays the admission of patients to the cohort area should be coordinated by the duty nursing officer.
- Housekeeping department is responsible for daily and terminal cleaning and audit the cleaning process in collaboration with ward nurses and ICP.

Definitions**Single-room isolation**

The most effective form of isolation is a single room with a self-contained toilet and its own hand basin. This should always be the first choice for placement of an infected/colonized patient. Bed management team is responsible for the identification of appropriate single room in the hospital.

Cohorting:

The practice of grouping patients infected or colonized with the same infectious agent together to confine their care to one area and prevent contact with susceptible patients (cohorting patients). During outbreaks, healthcare personnel may be assigned to a cohort of patients to further limit opportunities for transmission (cohorting staff).

Criteria for admitting patients to cohort area:

1. The patient is known to be colonized/infected with MDRO previously or detected in the current admission.
2. The patient doesn't have pending screening result for Candida auris or different MDRO.
3. The medical status of the patient is stable enough to be cared in cohort area.
4. The Patient is not on chemotherapy or heavy immunosuppressant.

Note: Patients tested MDRO positive two or more years ago and not admitted to any hospital after that positive result should not to be admitted in cohort ward unless the repeat MDRO screening is still positive.

Other cases can be considered for cohorting on individual basis after discussion with infection prevention & control team.

Procedure:

- Isolation should be initiated upon notification into a single room. If the number of MDRO cases exceeded the available single rooms then cohort should be considered when 2 or more patients having same organism (e.g. MRSA, MDRAB, VRE etc.).
- Patient who are known to be colonized/infected with mixed Gram Negative Bacilli can be cohorted.
- Consider cohorting patients into cubicle within wards if there is no dedicated ward available.
- Ensure, where possible, that care for cohorted patients is by designated staff that is not caring for other patients outside the cohort.
- Ensure effective isolation, i.e. cubicles backdoor should be closed all the time to provide physical separation from other patients. Daily cleaning is accomplished as delineated in this policy and report malpractices or drifting from following principles of infection prevention & control.
- During the working hours the ICP in-charge will be the focal point for coordination.
- After working hours, weekend and public holidays the duty nursing officer will liaise with the on-call ICP (5175).
- Contact Isolation sign on the entry of the cohort.
- Strict restrictions of visitors should be followed .Visitation permit for a single person should be issued to the first of kin only.
- Strict Contact isolation using the appropriate PPEand maintaining hand hygiene as per policy.
- Minimize the movement of patients for non-clinical reasons as much as possible.
- Use dedicated medical items/equipment for the cohort area. Limited supply of those should be kept in the cohort room.
- Dedicated toilet for the infected/colonized cases only and not to be shared by visitors and other patients.
- Dedicated cleaner and cleaning equipment.

Patient education

- Patients for possible cohort should be informed about cohort by responsible staff nurse or ward nurse.

- If attender is required, the responsible staff nurse should explain about importance of hand hygiene and maintaining contact precautions.

Visitors/attendants

- No attendants should be allowed with the patient unless it is extremely required.
- Nurse to inform about all patients who need attendants.
- Attendants who are not following the precautions should be counseled by the nurse in charge. Involve public relation officer (PRO) to talk to them.
- Infection Prevention & control team should be informed about non-compliant attendant to be counseled by ICP.
- Visitors should be restricted to close relatives only. Encourage relatives to enter isolation area one at a time wearing PPE and to perform Hand hygiene.

Refusal of patients to cohort

- If the patient is decided to go to cohort area and he /she or next of kin refuses to be transferred to the cohort, a counseling session should be conducted by the caring team Doctor and the nurse. Also inform and involve the IP&C team during working hours.
- In case counseling fails to convince the patient, admission to a single room should be arranged on a paying basis.

Closure of cohort

- Closure of cohort area should be undertaken after consultation with IP&C department.

Patients clothing and soiled laundry

All items of patient clothing that is either contaminated (i.e. soiled) or from a patient with an infection must be bagged in the red laundry bag with alginate (water soluble) strip designated for personal patient clothing.

Waste disposal

- All dust bins in the patient cohort area should be lined with yellow bags.
- Dustbins outside the patient area used for domestic waste should be lined with black bags.
- Sharp boxes should be available in all patient areas and handled according to IP&C policy.
- Cleaners should collect waste bags at least twice daily.

Routine daily room cleaning

- Enhanced environmental cleaning should be performed on every shift to reduce the bacterial load in the cohort area.
- High touch surfaces should be cleaned with hycoline as listed below:

Bed rails / controls	Tray table
IV pole (grab area)	Call box / button
Telephone	Bedside table handle
Chairs	Room sink
Room light switch	Room inner door knob

Terminal cleaning of cohort cubicle

- Nursing staff must inform the responsible housekeeping team & IP&C that the cubicle will need to be terminally cleaned if all patients are discharged or transferred elsewhere.
- The rest of the procedure to be followed according to terminal cleaning policy SOP/IPC/01/Vers.1.0

Daily Cleaning washroom and toilet assigned for the cohort

Daily cleaning with hycoline should carry preferably each shift to the following areas: Toilet door knobs, sinks, toilet seat, toilet handle, bathroom hand rails.
Use separate cloth for the handles, another one for toilet seat and another one for the sink.

Terminal Cleaning post patients' discharge/transfer

For terminal cleaning please refer to Terminal Cleaning Policy.

Fumigation

- Whenever possible fumigation of cubicle with proper sealing should be done after the terminal cleaning.

Audit

- IP&C team should audit compliance with policy.
- The checklist should be utilized by ICPs and Ward nurses.
- Terminal cleaning will be audited by housekeeping supervisor with IP&C team.
- Environmental audit should be done at least once per year by ICP and ward nurse.

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
GCC infection prevention & control Manual	GCC IC team	2013	51-53,69,80
Management of Multidrug-Resistant Organisms In Healthcare Settings, 2006	CDC, USA	November 13, 2015	24-29,45-46, 49
2007 Guideline for Isolation Precautions: Preventing Transmission of Infectious Agents in Healthcare Settings	CDC, USA	2007	57-58, 70

Review / Revision History

Revision History			
Revision	Description of change		Revision Date
1	Initial Release	Zaina AL Maskari	22/06/2016
2	Reviewed by	Jayalakshmi	22/02/2019
3	Reviewed by	Dr. Amal Al Tai	07/03/2022
4	Reviewed by	Ms. Khadija Al Zadjali	17/03/2024
Written by		Checked by	Authorized by
Zaina AL Maskari		Dr. Omaira Al Omairi	Dr. Amal Al Tai

Introduction

Healthcare associated infections are a major issue in the quality and safety of health care, in both hospital and community settings. Identification of patients with specific multi drug resistant organisms (MDRO's), can reduce the risk of transmission of these organisms within healthcare and thus reduce morbidity and mortality associated with these organisms.

1. Purpose:

This policy provides instruction to staff on indication & sampling procedure for MDRO's and candida auris screening. The current MDRO of concerns in our hospital are the following:

MDRA: Multi-Drug Resistant *Acinetobacter Baumannii*

CRE: Carbapenem Resistant *Enterobacterales*

MRSA: Methicillin Resistant *Staphylococcus aureus*

VRE: Vancomycin Resistant *Enterococci*

MDR Pseudomonas: Multidrug Resistant *Pseudomonas*

ESBL: Extended Spectrum beta Lactamase

The principle objectives of screening for MDRO and candida auris are:

1. To identify patients who are carriers of MDRO and candida auris.
2. To reduce the risk of patients colonized with MDRO and candida auris developing infection
3. To reduce the risk of transmission of MDRO and candida auris to other patients.

2. Scope:

This policy applies to all patients admitted to Royal hospital. The policy should be applied by all staff working at royal hospital.

3. Role and responsibilities:

➤ The Infection prevention & Control (IP&C) committee:

The committee is responsible to oversee the ongoing development of this policy

➤ Head of IP&C department:

Head of IP&C is responsible for overseeing the implementation of the policy, for reporting any concerns and performance to the administration and/or to infection prevention & control committee.

➤ Infection Preventionist (IP's):

IP's are responsible for training, supporting implementation of the policy and monitoring compliance with the policy.

➤ Microbiology laboratory:

The microbiology lab is responsible for releasing the results of all MDRO screening samples. In case of suspicion of an outbreak the head of IP&C must be notified by the lab.

➤ Nursing staff:

Nurses are responsible for ensuring adherence of all the staff to this policy, collecting samples as outlined in this policy and initiate the isolation/precautions as outlined in the isolation policy for suspected and confirmed cases.

➤ Medical staff:

Doctors are responsible to inform their patients in case the screening come positive and educate them.

4. Definitions

➤ **MDRO:** a bacterial strain that is resistant to one or more of antimicrobial agents.

➤ **MDRAB:**

Any *Acinetobacter spp.* testing non-susceptible (i.e., resistant or intermediate) to at least one agent in at least 3 antimicrobial classes of the following 6 classes:

1. Ampicillin/sulbactam
2. Cephalosporins (cefepime, ceftazidime)
3. β -lactam/ β -lactam β -lactamase inhibitor combination (piperacillin, piperacillin/tazobactam)
4. Carbapenems (imipenem, meropenem, doripenem)
5. Fluoroquinolones (ciprofloxacin or levofloxacin)
6. Aminoglycosides (gentamicin, tobramycin, or amikacin).

In A'Shifa the alert appears for positive cases MDRA

➤ **CRE:** Any Enterobacterales e.g. *Escherichia coli*, *Klebsiella oxytoca*, *Klebsiella pneumoniae*, or *Enterobacter spp.* testing resistant to imipenem, meropenem, doripenem, or ertapenem by standard susceptibility testing methods (i.e., minimum inhibitory concentrations of ≥ 4 mcg/mL for doripenem, imipenem and meropenem or ≥ 2 mcg/mL for ertapenem) OR by production of a carbapenemase (i.e., KPC, NDM, VIM, IMP, OXA-48) demonstrated using a recognized test (e.g., polymerase chain reaction, metallo- β -lactamase test, modified-Hodge test, Carba-NP). In A'Shifa the alert appears for positive cases **CRE**

- **MRSA:** Includes *S. aureus* cultured from any specimen that tests oxacillin-resistant, cefoxitin-resistant, or methicillin-resistant by standard susceptibility testing methods, or by a laboratory test that is FDA-approved for MRSA detection from isolated colonies; these methods may also include a positive result by any FDA-approved test for MRSA detection from specific sources. In A'Shifa the alert appears for positive cases **MRSA**
- **VRE:** *Enterococcus faecalis*, *Enterococcus faecium*, or *Enterococcus species unspecified* (only those not identified to the species level) that is resistant to vancomycin, by standard susceptibility testing methods or by results from any FDA-approved test for VRE detection from specific specimen sources. In A'Shifa the alert appears for positive cases **VRE**
- **MDR-Pseudomonas:** Any *Pseudomonas* spp. testing non-susceptible (i.e., resistant or intermediate) to at least one agent in at least 3 antimicrobial classes of the following 5 classes:
 - Cephalosporins (cefepime, ceftazidime)
 - β -lactam/ β -lactam β -lactamase inhibitor combination (piperacillin, piperacillin/tazobactam)
 - Carbapenems (imipenem, meropenem, doripenem)
 - Fluoroquinolones (ciprofloxacin or levofloxacin)
 - Aminoglycosides (gentamicin, tobramycin, or amikacin)

In A'Shifa the alert appears for positive cases **PAER**

➤ **ESBL:**

Are enzymes that mediate resistance to extended-spectrum (third generation) cephalosporins (e.g., ceftazidime, cefotaxime, and ceftriaxone) and monobactams (e.g., aztreonam) but do not affect cephamycins (e.g., cefoxitin and cefotetan) or carbapenems (e.g., meropenem or imipenem).

In A'Shifa the alert appears for positive cases **ESBL**.

- **Candida Auris:** an emerging fungus that is resistant to multiple classes of antifungal medication (eg: azoles, amphotericin B, echonocandins).
- **MDROs colonization risk factors:** presence of tracheostomy, urinary catheter, chronic wound
- **Colonization** means the MDRO are present in or on the body but are not causing illness. Healthy people may carry the bacteria causing MDRO without becoming ill.
- **Infection** means the MDRO are present in or on the body and are causing illness. Symptoms of MDRO may vary depending on the part of the body that is infected. Infection can occur in any part of the body. Infections may affect the skin, lungs, urinary tract or bloodstream

5. MDROs included in the active surveillance

Active Surveillance	Rescreen from the same site if positive from clinical samples
MRSA, CRE, MDRAB, VRE and Candida auris	ESBL, MDR Pseudomonas

The active surveillance means screening actively to detect carriers. For MDRO that are not included in the active surveillance if detected in clinical samples the patient is screened from the same site for the purpose of discontinuing isolation only.

PAER screening:

There is no active surveillance screening for Multidrug resistant Pseudomonas aeruginosa (PEAR). Rescreening is required for de-isolation purpose only. When the sample is collected for PEAR screening, it is important to specify this in the remark section. Contact Infection prevention & control staff for further advice.

ESBL screening:

There is active screening for ESBL in the NICU only. Patients colonized or infected with ESBL are isolated in the NICU and surgical areas. ESBL rescreening is required for de-isolation of patients at these areas. When the sample is collected for ESBL screening, it is important to specify this in the remark section. Contact Infection prevention & control staff for further advice

6. Screening purposes:

7.1 Screening patients at the time of admission:

For patients to be transferred directly to Royal Hospital the following should be followed:

- If the case is urgent it can be accepted if his/her screening results are not known and the screening is initiated at Royal Hospital. The patient should be under contact precautions till the results of screening tests released. If results are positive, continue contact precautions as per isolation policy
- For non-urgent transfer, enquiring about the screening results is a must before accepting the patient, accordingly the bed can be arranged as for isolation or not, the screening samples has to be taken within 48 hours before transferring.
- If the screening results are not available within 48hours before transfer, screening to be repeated.
- If results from referring hospital is negative standard precautions should be applied. If results are positive, ask the referring hospital to attach the result in the referring notes.

The following table (1) lists the indications and samples for screening patients based on risk factors.

Type of organism	Indications for screening	Screening request and samples
MDRA/ CRE	1. Admission abroad within the last 6 months 2. Admission to other local hospital within the last 3 months 3. Readmission to RH with risk factors e.g. wounds, tracheostomy, catheter, central line 4. Admission to critical unit (AICU, PICU, HD) 5. Admission to post cardiac surgery unit after more 5 days stay in other wards	<u>CRE/MDRA screening culture</u> Rectal swab for body screening Urine, swab or respiratory sample according to the risk factors
VRE	1. Admission abroad within the last 6 months 2. Admission to other local hospital within the last 3 months 3. Readmission to RH with risk factors e.g. wounds, tracheostomy, catheter, central line 4. Admission to critical unit (AICU, PICU, HD) 5. Admission to post cardiac surgery unit after more 5 days stay in other wards 6. Monthly screening for chronic oncology, hematology and hemodialysis patients	<u>VRE screening culture</u> Rectal swab for body screening Urine, swab or respiratory sample according to the risk factors

MRSA	1. Admission abroad within the last 6 months 2. Admission to other local hospital within the last 3 months 3. Readmission to RH with risk factors e.g. wounds, tracheostomy, catheter, central line 4. Admission to critical unit (AICU, PICU, HD) 5. Admission to post cardiac surgery unit after more 5 days stay in other wards 6. Prior to cardiac surgeries and transplant surgeries (renal and liver). 7. Prior to insertion of dialysis line (hemodialysis or peritoneal dialysis)	<u>MRSA screening culture</u> Nasal, axilla and groin swabs for body screening Urine, swab or respiratory sample according to the risk factors <u>MRSA PCR (Prior to urgent cardiac surgery)</u> Nasal PCR swab for MRSA PCR
Candida Auris	1. Admission abroad within the last 6 months 2. Adult patient with recent admission to other local hospital within the last 3 months 3. Pediatric patients with recent admission to other local hospital endemic with Candida auris within the last 3 months 4. Admission to adult critical units	<u>Canidia auris screening</u> Axilla and groin swabs for body screening Urine, swab or respiratory sample according to the risk factors

7.2 Screening patient admitted to NICU:

The patients admitted to the NICU are screened for CRE, MDRA, MRSA, ESBL Klebsiella and Serratia if meeting the following criteria.

- Baby or mother was admitted to other hospital in the last 3 months
- Unknown/ abandoned baby
- If mother tests positive for an MDRO, screen the baby for the same MDRO.

The screening samples for CRE, MDRA and MRSA are as described above.

The screening samples for ESBL/Serratia are:

- Rectal swab for body screening
- Urine, swab or respiratory sample according to the risk factors

7.3 Screening patient for the purpose of deisolation:

MDROs screening for the purpose of deisolation has to be done while patient is not on antimicrobials. Follow the following algorithms for the purpose of de-isolation:

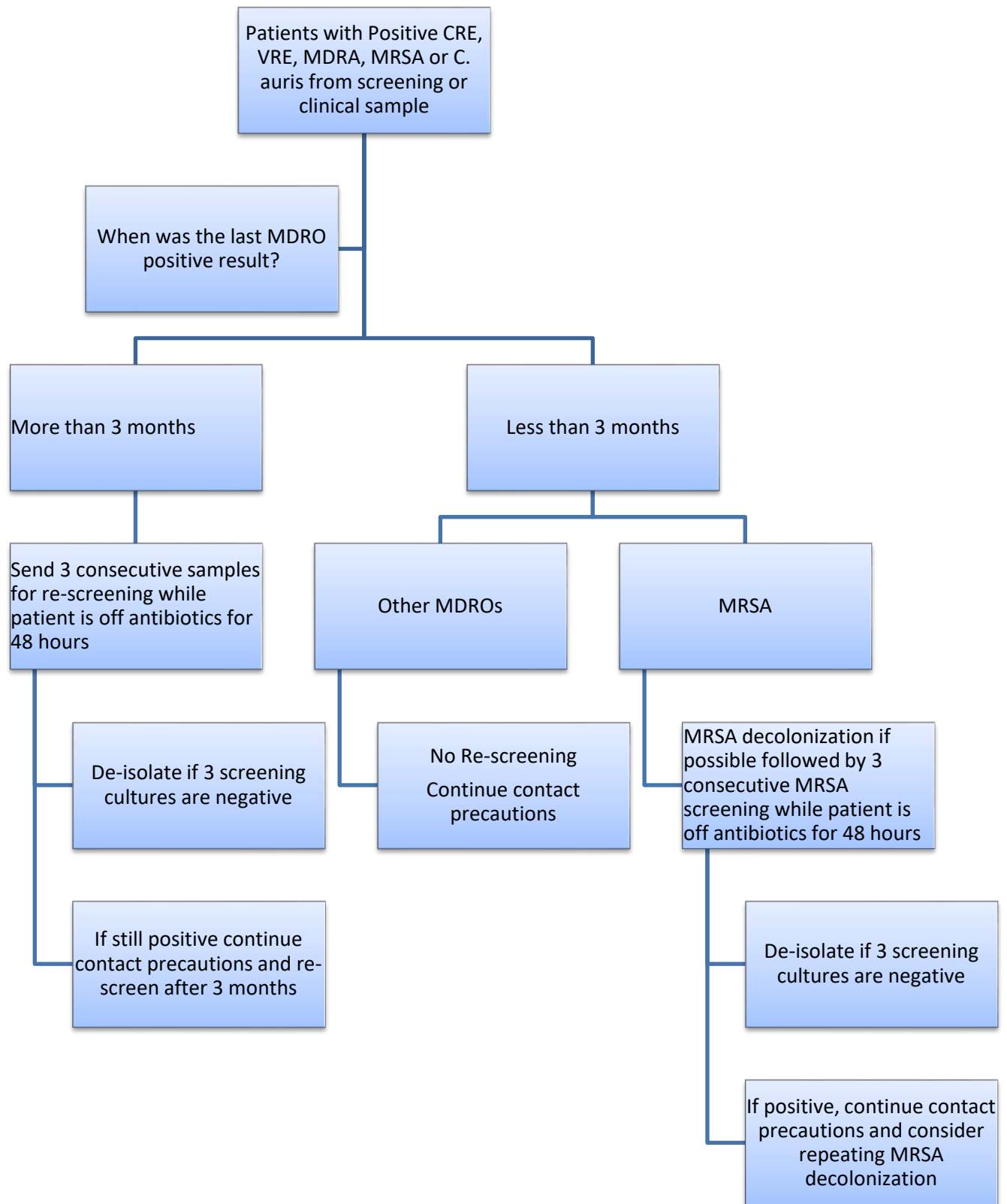


Figure 1: De-isolation of CRE, VRE, MDRA, MRSA and Candida auris

If the patient had an ESBL in the surgical areas or MDR *Pseudomonas aeruginosa* in any area, apply the same isolation procedures for other MDROs. Follow the following algorithms for de-isolation:

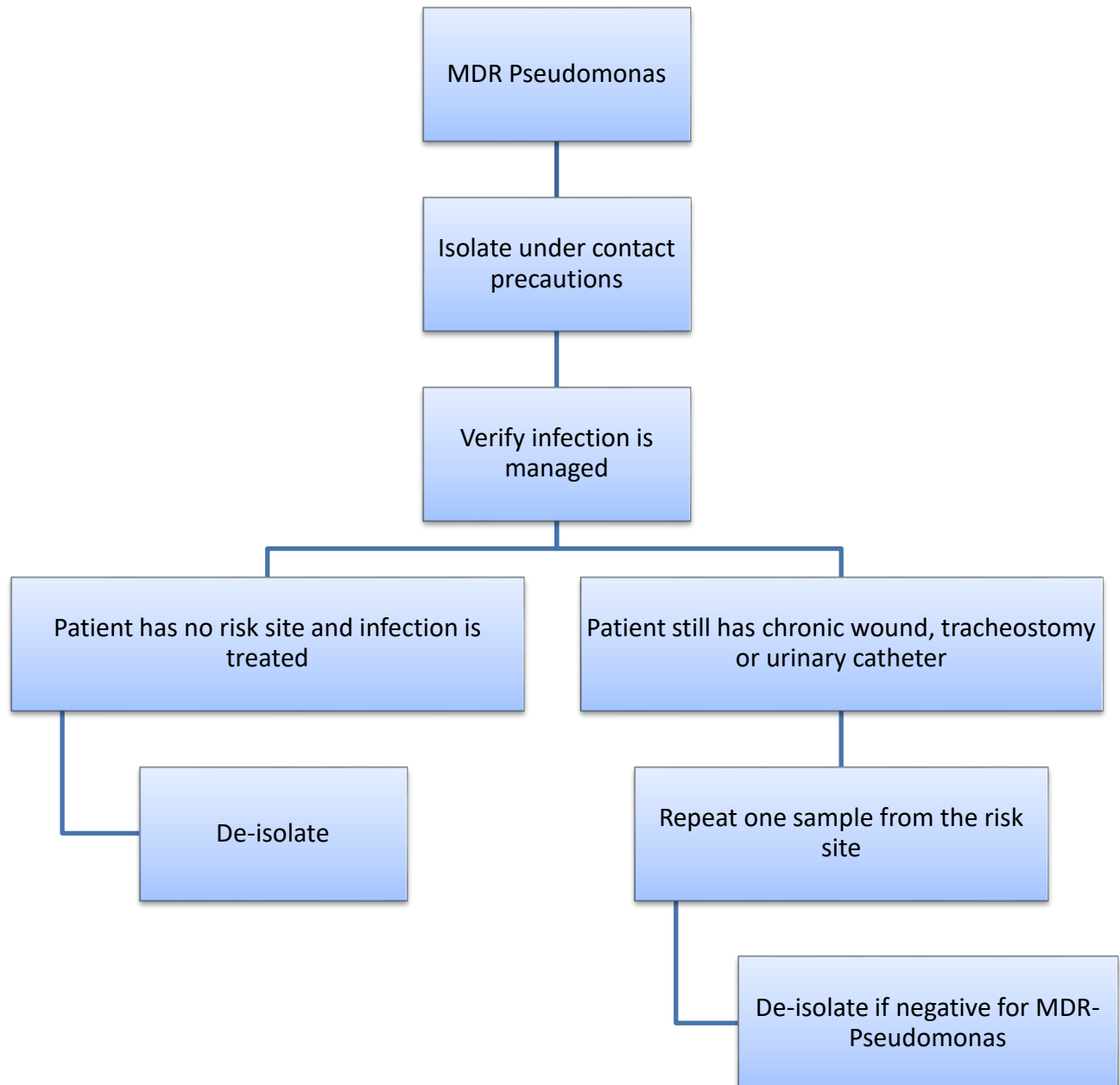


Figure 2: De-isolation of MDR *Pseudomonas aeruginosa*

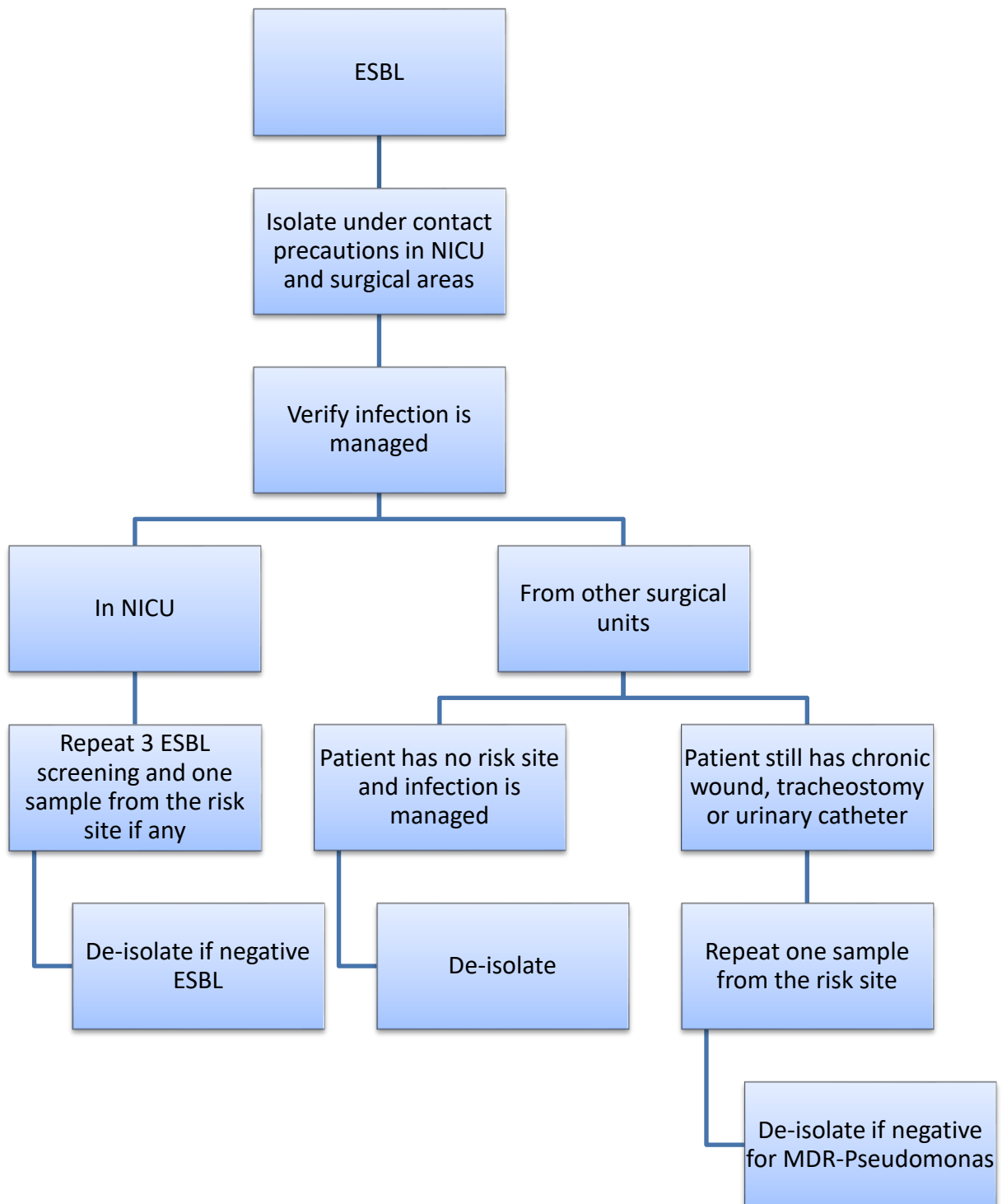


Figure 3 : De-isolation of patients with ESBL organisms. Note that the same algorithm is applicable for Serratia in the NICU.

7.4 Screening patient for the purpose of shifting to other hospital:

Patients can be screened before shifting if requested by the receiving hospital. In case the patient is on antibiotics inform the receiving hospital that screening was done while patient was on antibiotics.

7.5 Contact screening:

- **MDRO Contact**
- Screening of contacts is only required if the index patient is MDRO positive and the patient has not been isolated on admission, or during an outbreak situation.
- A patient who has shared the same bed area for more than 24 hours with MDRO colonised or infected patient where contact precaution were not in place
- Screening of patients in the same setting is not normally required if the case was identified on admission and the patient isolated immediately
- Screening of household contacts and healthcare staff is not required. There is no compelling evidence to suggest that screening the household contacts or healthcare workers to check for colonisation will provide additional benefit in controlling spread in the healthcare setting

7.6 Screening during an outbreak:

As advised by outbreak management team & IP&C department depending on the micro-organism involved.

7. Specimen Collection Procedure:

8.1 General instructions:

- Please note that privacy and comfort should be assured before commencing sample collection.
- Explain to the patient the procedure before initiating the screening
- Before collection hand hygiene should be performed and then disposable gloves should be donned.
- Pre-moisten all swabs with sterile normal saline prior to taking a specimen.
- Swab wounds, skin lesions, incisions, ulcers or central line exit site if present with separate swabs.
- All swabs collected should correctly labelled with patient sticker or hospital number, screened site and MDRO Code (MRSA, CRE, MDRAB, VRE, etc.) for screening
- Each MDRO of interest should have a separate swab collected.
- Please note that of all body sites being swabbed the rectum should always be the last in the swabbing sequence.

8.2 Collection Procedure

8.2.1 Nasal swab

- Open and remove sterile tipped swab applicator from transport casing and pre-moisten it.
- Taking care to avoid other contact with swab, insert the swab approximately 1-2 cm (approximately $\frac{3}{4}$ inches) into the first nostril next to the nasal septum.
- Rotate the swab against the anterior nasal mucosa for 3 – 5 seconds.
- Using the same swab, repeat for the other nostril.
- Carefully place used swab back into transport tube and secure.

8.2.2 Axilla & groin swabs:

- Clean the site with normal saline to remove debris before swabbing.
- Open and remove sterile tipped swab applicator from transport casing and pre-moisten it.
- Use the same swab for identical sites: one swab for both axilla and one swab for both inguinal areas.

8.2.3 Rectal swab

- Ask the patient to loosen their clothing
- Open and remove sterile tipped swab applicator from transport casing and pre-moisten it.
- Taking care to avoid other contact with swab, rotate the swab against the external rectal orifice. If visible stool is not obtained on the swab, insert it a few millimetres into the rectum until visible stool is obtained.
- Carefully place used swab back into transport tube and secure.

8.2.4 Other samples:

For sample collection from other risk sites such as wounds, ET secretion and urine follow microbiology laboratory guidance.

8. Acting on Results of Screening / Isolation of Patients:

Any patient screened for MDROs and results became positive follow isolations policy for further actions.

9. References:

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
Screening, Testing and Surveillance for Antibiotic-Resistant Organisms (AROs) in healthcare settings	Provincial Infectious Diseases Advisory Committee	2.2013	7, 40-47
Methicillin Resistant Staphylococcus Aureus (MRSA) Protocol	King's College Hospital NHS trust, UK	9.2011	4-5,7
Protocol for CRA MRSA Screening National Rollout in Scotland	Health Protection Scotland	31.1.2013	3-4, 11-15
GCC infection prevention & control Manual	GCC Center of IC	2013	108-109

10.Review / Revision History:

Revision History			
Revision	Description of change	Author	Revision Date
1	Initial Release	Amina AL Jardani	16/11/2010
2	Reviewed	Zaina AL Maskari	16/11/2016
3	Reviewed	Ahlam Al Baloushi Khadija AlZadjali	20/04/2022
4	Reviewed	Maryam Al Balushi	12/03/2023
5	Changed the criteria for MDRO screening in the NICU. Added algorithm for de-isolation of ESBL and MDR pseudomonas	Omaira Al Omairi	4/11/2024
Written by		Checked by	Authorized by
Zaina AL Maskari		Dr. Omaira Al Omairi	Dr. Amal Al Tai

	Sultanate of Oman Ministry of Health The Royal Hospital Department of Infection Prevention & Control	Infection Prevention & Control policy
		Page of
POL/IPC/Ver.1.0	Transmission based precautions	
Effective Date: Target Review Date:	Version 1.0	

1.0 INTRODUCTION

The infection prevention and control measures are designed to stop the spread of infectious diseases by breaking the chain of transmission. To stop the transmission of some infectious pathogens the standard infection control precautions have to be combined with extra precautions depending on the mode of transmission of those organisms i.e. transmission-based precautions.

2.0 PURPOSE

This document aims to provide a practical guidance on the isolation precautions to minimise the risk of transmission of epidemiologically important pathogens during the admission of patient.

3.0 SCOPE

This document applies to ALL healthcare facilities, patients and visitors who have potential exposure to infectious materials, including body substances, contaminated environmental surfaces or contaminated air.

4.0 DEFINITIONS

TRANSMISSION-BASED PRECAUTIONS are sets of measures designed for the care of patients who are **KNOWN OR SUSPECTED** to be infected/colonized by epidemiologically important or highly transmissible organisms/infections.

Transmission-Based Precautions are categorized into Contact Precautions, Droplet Precautions, Airborne Precautions, Enteric precaution, according to the predominant mode(s) of transmission of infectious agents. Refer to appendix A for examples of each type.

5.0 PROCEDURES

5.1 Contact precautions

5.1.1 Contact precautions indications:

Contact precautions are designed to prevent transmission of microorganisms by direct or indirect contact with the patient or patient environment. They are used for patients with known or suspected to be colonized with MDROs. They are also applicable for patients who have excessive wound discharge, faecal incontinence or other discharges from the body which has increase potential for environment contamination. Other indications for contact precautions are scabies and head/ body lice.

5.1.2 Contact precautions Procedure:

- Patient placement:
 - Explain the need for the precautions to the patient.
 - Preferably, place the patient in a single room with ensuite bathroom.
 - If not possible, cohort patients infected or colonized with the same microorganism and ensure physical separation of patients and assign a designated bathroom for them. The decision to cohort should be done in conjunction with the infection prevention and control staff.
 - Place contact precautions signage at entrance of patient's zone (appendix B) either on the outside of the door or the curtain.
- Patient care equipment and devices:
 - Use dedicated non-critical patient care equipment for a single patient if possible. Otherwise, clean and disinfect the equipment according to manufacturer's cleaning instructions before using on another patient.
 - Keep essential supplies to minimum inside the room.
- Personal protective equipment (PPE):
 - Use appropriate PPE upon room entry and discard before exiting patient room
 - Wear a gown and gloves for all interactions that may involve contact with patient or patient environment
 - Mask and Eye cover to be used if needed e.g. suspected aerosols generation.
- Hand hygiene:
 - Avoid unnecessary touching of the patient environment
 - Perform hand hygiene according to the standard precautions
 - Do not forget hand hygiene after taking off gloves

- Linen management:
 - Line the hamper with a water-soluble red bag and keep inside the patient room.
- Environmental cleaning:
 - Ensure consistent cleaning and disinfection of surfaces in close proximity to the patient and those likely to be touched by the patient and healthcare workers.
 - When possible, use 'No-Touch Disinfection' (such as UV disinfection, H₂O₂-based disinfection) in addition to manual cleaning and disinfection to enhance the effectiveness of the environmental cleaning and disinfection.
- Patient transfer:
 - Minimize patient transfer as possible
 - If the patient has to go to a different department, inform the receiving department /team of the patient's transmission-based precaution so they can take the necessary precautionary steps to prevent spread of disease.
 - When transporting patient outside of the room remove PPE and perform hand hygiene, as above, after placing patient on trolley/stretcher/wheelchair

5.2 Droplet precaution

5.2.1 Droplet precautions indications:

Apply droplet precautions for patients confirmed or suspected to be infected with pathogens that are transmitted by large droplet particles >5µm in size. The common indications are respiratory viruses, pertussis, mumps, Rubella, Parvovirus, Diphtheria (pharyngeal), group A streptococcus, Haemophilus influenza, Nisseria meningitidis & Epiglottitis.

5.2.2 Droplet precautions Procedure:

- Patient placement:
 - Explain the need for the precautions to the patient.
 - Preferably, place the patient in a single room with ensuite bathroom.
 - If not possible, cohort patients infected with the same microorganism and ensure physical separation of patients (at least 2m). The decision to cohort should be done in conjunction with the infection prevention and control staff.
 - Place droplets precautions signage at entrance of patient's zone (appendix C) either on the outside of the door or the curtain.
- Patient care equipment and devices:
 - Use dedicated non-critical patient care equipment for a single patient if possible. Otherwise, clean and disinfect the equipment according to manufacturer's cleaning instructions before using on another patient.
- Personal protective equipment (PPE):

- Use appropriate PPE (Mask, Gown and gloves) upon room entry and discard before exiting patient room
- Masks should be removed and disposed of on leaving the patient's zone (e.g. at the door, curtain or the anteroom) and perform hand hygiene
- Hand hygiene:
 - Avoid unnecessary touching of the patient environment
 - Perform hand hygiene according to the standard precautions
 - Do not forget hand hygiene after taking off gloves
- Linen management:
 - Line the hamper with a water-soluble red bag and keep inside the patient room.
- Environmental cleaning:
 - Ensure consistent cleaning and disinfection of surfaces in close proximity to the patient and those likely to be touched by the patient and healthcare workers.
 - Depending on the microorganism, disinfection may be required in addition to cleaning.
- Patient transfer:
 - Minimize patient transfer as possible
 - If the patient has to go to a different department, inform the receiving department /team of the patient's transmission-based precaution so they can take the necessary precautionary steps to prevent spread of disease.
 - If clinically able, patient should wear surgical mask and follow respiratory hygiene/cough etiquette when outside of the usual patient zone

5.3 Airborne precautions

5.3.1 Airborne precautions indications:

Apply Airborne Precautions for patients documented or suspected to be infected with pathogens that are transmitted by airborne droplet nuclei $< 5\mu\text{m}$ in size. The commonest indication for airborne precautions (Appendix D) is tuberculosis (pulmonary, laryngeal and extra-pulmonary with draining lesions). Some viral infections require airborne and contact precautions e.g. Chickenpox, disseminated herpes zoster, measles and viral haemorrhagic fevers. Refer to Appendix E.

5.3.2 Airborne precautions procedures:

- Patient placement:
 - Explain the need for the precautions to the patient.

- Place the patient in a negatively pressurised single room with ensuite bathroom. Discharge of air to the outside or HEPA-filtered before recirculation.
 - If isolation in a negative pressure room is not available, place the patient in a single room where the door should be closed at all times.
 - Place airborne (Appendix D) or airborne/contact (appendix E) precautions signage at entrance of patient's zone (appendix E).
- Entry restrictions:
 - Restrict susceptible healthcare personnel and visitors (e.g. pregnant, immunocompromised, non-immune) from entering the rooms
- Personal protective equipment (PPE):
 - Wear fit tested N95 mask (appendix
 - If patient needs contact precautions in addition to the airborne precautions, use other appropriate PPE upon room entry and discard before exiting patient room (Appendix E)
- Hand hygiene:
 - Perform hand hygiene according to the standard precautions
- Environmental cleaning:
 - Ensure consistent cleaning and disinfection of surfaces in close proximity to the patient and those likely to be touched by the patient and healthcare workers.
 - Depending on the microorganism, disinfection may be required in addition to cleaning.
- Patient transfer:
 - Minimize patient transfer as possible
 - If the patient has to go to a different department, inform the receiving department /team of the patient's transmission-based precaution so they can take the necessary precautionary steps to prevent spread of disease.
 - Patient should wear surgical mask and follow respiratory hygiene/cough etiquette when outside of the usual patient zone

5.4 Enteric precautions

5.4.1 Enteric precaution indications:

Enteric precaution is used for the patients with known or suspected gastroenteritis or infectious diarrhea. It's used to prevent infections that are transmitted by direct or indirect contact with fecal materials. Refer to appendix F.

5.4.2 Enteric precaution procedures:

- Hand hygiene has to be performed using soap and water NOT hand rub.
- Use sporicidal disinfectants when cleaning rooms housing patients infected or colonized by spore-forming organisms such as *Clostridium difficile*.
- Other procedures for enteric precautions are the same like contact precautions

5.5 Precautions for Family/Visitors to Patients under Transmission-Based Precautions

- All visitors should perform hand hygiene prior to entering a patient room and immediately after leaving the room.
- For parents/visitors with extended stay in a patient's room, including overnight visitation, Transmission-Based Precautions must be implemented

Types of Transmission	EXAMPLES of conditions or infectious agents
Contact Transmission	-Multi-Drug Resistant organisms (MDROs) e.g. <i>Acinetobacter baumannii</i> (MDRAB) -Extended Spectrum Beta-Lactamases (ESBL) in surgical wards and NICU only -Carbapenem-Resistant Enterobacteriaceae (CRE) -Vancomycin Resistant Enterococci (VRE) -Methicillin-Resistant <i>Staphylococcus aureus</i> (MRSA) -Scabies - Hepatitis B, C or HIV with active bleeding - Ebola Virus

Droplet Transmission	-Bordetella pertussis (Whooping cough) - Respiratory virus e.g. Influenza virus, Adenovirus, Rhinovirus, RSV -Mycoplasma pneumonia -SARS-associated coronavirus including SARSCoV 2 -Group A Streptococcus - Neisseria meningitides -Rubella - Mumps
Airborne Transmission	-Pulmonary or laryngeal tuberculosis (TB) -Measles -Varicella / Herpes zoster (disseminated) -Novel / emerging respiratory illnesses -Smallpox (Variola virus) - SARS CoV2 infection with Aerosol Generating Procedures (AGPs) such as suctioning, endoscopy, bronchoscopy, nebulization, sputum induction, endotracheal intubation
Enteric Transmission	-Clostridium difficile -Norovirus - Rotavirus - Hepatitis A or E with diarrhoea -Acute diarrhoea

Appendix (A):Types of transmission-based precautions



CONTACT PRECAUTIONS



Families and visitors follow isolation instructions from information sheet
If you have questions go to nurse station
أعزائي الأقارب والزوار الرجاء إتباع إرشادات العزل المذكورة . إذا كان لديك أي
استفسار الرجاء التوجه للممرضة المسؤولة

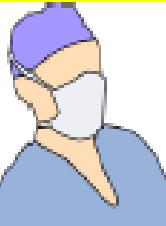


Clean hands when entering and leaving the room.
Practice the 5 moments of hand hygiene.
تنظف يديك قبل وبعد دخول غرفة العزل



Wear Gown and Gloves at door
إلبس الرداء الواقي والثغافازات قبل دخول غرفة العزل





Wear Mask and eye cover if there is splashing of body fluids
is likely
إلبس القناع وواقي العين في حالة التعرض للرشاذا والسوائل



Use patient dedicated or disposable equipment
Clean and disinfect shared equipment
استخدم الادوات المخصصة للمريض في غرفة العزل
تنظف وحقم الادوات بين كل مريض وآخر



Royal Hospital
Infection prevention &Control

CONTACT PRECAUTIONS

Display sign outside the door, Remove sign after discontinuation of isolations

Common conditions:

Multi-Drug Resistant Organisms (MDRA, MRSA, VRE, CRE, ESBL, PAER & others)

Scabies

Draining and uncontained wounds and abscesses

Herpes Zoster (localized)

Disseminated and neonatal Herpes Simplex infection

Burkholderia cepacia in Cystic fibrosis patient, including colonization

Head/body lice

Any other conditions as recommended by Infection Control

Dishes/Utensils:

No special precautions. Kitchenware sanitized in dishwasher

Equipment and Supplies:

Use dedicated or disposable equipment when available

Clean and disinfect reusable equipment including: IV pumps, cell phone, pagers (if used in room) and other electronics, supplies and equipment prior to removing from patient's room.

Keep essential supplies to the minimum inside the room.

Linen Management:

Line the hamper with a water-soluble red bag and keep in the patient room

Personal Protective Equipment (PPE):

Don PPE upon entry

Discard before exiting the patient room in the yellow bag.

<u>Put ON in this order</u>	<u>Take OFF in this order</u>
1. Wash or gel hands	1. Gloves
2. Gown	2. Wash or gel hands (even if gloves used)
3. Mask (only if needed)	3. Eye cover (if used)
4. Eye cover (if needed)	4. Gown
5. Gloves	5. Wash or gel hands
	6. Mask (if used)
	7. Wash or gel hands

Private Room

IF not available. Cohort patient that has the same organism but no other infection

Room Cleaning:

Daily cleaning of high touch surfaces

Appendix B

Families and visitors follow isolation instructions from information sheet

If you have questions go to nurse station

أعزائي الاقارب والزوار الرجاء إتباع إرشادات العزل المذكورة . إذا كان لديك أي
استفسار الرجاء التوجه للممرضة المسؤولة



Clean hands when entering and leaving the room.
Practice the 5 moments of hand hygiene.

نظف يديك قبل دخول غرفة العزل



Wear Gown and Gloves at door

إلبس الرداء الواقي والقفازات قبل دخول غرفة العزل



Wear Mask and

إلبس القناع الواقي



Wear eye cover if there is splashing of body fluids is likely

إلبس واقى العين في حالة التعرض للرداذ والسوائل

DROPLET PRECAUTIONS

Display sign outside the door, Remove sign after discontinuation of isolations

Common conditions:

Mumps	Plague (Pneumonic)
Epiglottitis	Influenza & other respiratory viruses
Pertussis	Parvovirus
Diphtheria (Pharyngeal)	Group A Streptococci
<i>Neisseria meningitidis</i>	<i>Haemophilus influenzae</i>
Rubella	

Any other conditions as recommended by Infection Control

Dishes/Utensils:

No special precautions. Kitchenware sanitized in dishwasher

Equipment and Supplies:

Use dedicated or disposable equipment when available
Clean and disinfect reusable equipment including: IV pumps, cell phone, pagers (if used in room) and other electronics, supplies and equipment prior to removing from patient's room.
Keep essential supplies to the minimum inside the room.

Linen Management:

Line the hamper with a water-soluble red bag and keep in the patient room

Personal Protective Equipment (PPE):

Don PPE upon entry

Discard before exiting the patient room in the yellow bag.

Put ON in this order

1. Wash or gel hands
2. Gown
3. Mask
4. Eye cover (if needed)
5. Gloves

Take OFF in this order

1. Gloves
2. Wash or gel hands (even if gloves used)
3. Eye cover (if used)
4. Gown
5. Wash or gel hands
6. Mask
7. Wash or gel hands

Private Room

IF not available. Cohort patient that has the same organism but no other infection

Room Cleaning:

Daily cleaning of high touch surfaces

Terminal cleaning procedure, including cubicle curtain changes

Transport:

Essential transport only.

Alert receiving department regarding patient isolation status.

Ask the patient to wear a surgical mask & to follow respiratory hygiene

Clean and disinfect transport vehicle.

Discontinuation of isolation:

Discontinue precautions as per isolation policy or infection control instructions

Appendix (C): Droplet precautions

	AIRBORNE PRECAUTIONS Families and visitors follow <u>isolation</u> instructions from information sheet If you have questions go to nurse station أعزائي الأقارب والزوار الرجاء إتباع إرشادات <u>العزل</u> المذكورة . إذا كان لديك أي استفسار الرجاء التوجه للممرضة المسؤولة	
	Clean hands when entering and leaving the room. Practice the 5 moments of hand hygiene. نظف يديك قبل دخول غرفة العزل ، اتبع الحالات الخمس الهامة لنظافة الأيدي	
	Wear N95 Mask prior to entering the room الغرفة ليس القناع الواقي المختص N95 قبل دخول	
	Negative pressure isolation room is required Keep door always closed يحتاج غرفة سالبة الضغط أبقي الباب مغلقا دائما	
	Limit the movement/transport of patients to essential purposes only, During transport place surgical mask on the patient قلل من خروج المريض من الغرفة إلا للأسباب الضرورية ، في حالة نقل المريض بلبس المريض قناعا جراحيا واقيا	
	Royal Hospital Infection prevention &Control	

AIRBORNE PRECAUTIONS

Display sign outside the door, Remove sign after discontinuation of isolations

Common conditions:

Tuberculosis: pulmonary, laryngeal, and extra-pulmonary with draining lesions

Any other conditions as recommended by Infection Control

Dishes/Utensils:

No special precautions. Kitchenware sanitized in dishwasher

Equipment and Supplies:

Use dedicated or disposable equipment when available

Clean and disinfect reusable equipment including: IV pumps, cell phone, pagers (if used in room) and other electronics, supplies and equipment prior to removing from patient's room.

Keep essential supplies to the minimum inside the room.

Linen Management:

Line the hamper with a water-soluble red bag and keep in the patient room

Personal Protective Equipment (PPE):

Fit testing is required to determine size of N95 mask

Wear N95 mask upon entry

Remove the mask outside the room.

Put ON in this order

1. Wash or gel hands
2. N95 Mask

Take OFF in this order

1. N95 Mask
2. Wash or gel hands (even if gloves used)

Private Room

Isolate in a negative pressure room or neutral pressure single room, keep door closed all the time

Room Cleaning:

Daily cleaning of high touch surfaces

Terminal cleaning procedure, including curtain changes

Transport:

Essential transport only.

Alert receiving department regarding patient isolation status.

Ask the patient to wear a surgical mask & to follow respiratory hygiene

Clean and disinfect transport vehicle.

Appendix (D): Airborne precaution

	AIRBORNE/CONTACT PRECAUTIONS Families and visitors follow <u>isolation</u> instructions from information sheet If you have questions go to nurse station أعزائي الأقارب والزوار الرجاء إتباع إرشادات العزل المذكورة . إذا كان لديك أي إستفسار الرجاء التوجه للممرضة المسؤولة	
	Clean hands when entering and leaving the room. Practice the 5 moments of hand hygiene. نظف يديك قبل دخول غرفة العزل ، اتبع الحالات الخمس الهامة لنظافة الأيدي	
	Wear N95 Mask prior to entering the room لبس القناع الواقي المخصص N95 قبل دخول الغرفة	
	Wear Gown and Gloves at door لبس الرداء الواقي والقفازات قبل دخول غرفة العزل	
	Negative pressure isolation room is required Keep door always closed يحتاج غرفة سالبة الضغط، أبقى الباب مغلقا دائما	
	Limit the movement/transport of patients to essential purposes only, During transport place surgical mask on the patient قلل من خروج المريض من الغرفة إلا للأسباب الضرورية ، في حالة نقل المريض يلبس المريض قناعا جراحيا واقيا	
	Royal Hospital Infection prevention &Control	

AIRBORNE/CONTACT PRECAUTIONS

Display sign outside the door, Remove sign after discontinuation of isolations

Common conditions:

Measles

Viral Hemorrhagic Fever

Severe Acute Respiratory Syndrome (SARS)

Varicella infection:

Chickenpox, multi- dermatomal zoster and zoster in immunocompromised

Any other conditions as recommended by Infection Control

Dishes/Utensils:

No special precautions. Kitchenware sanitized in dishwasher

Equipment and Supplies:

Use dedicated or disposable equipment when available

Clean and disinfect reusable equipment including: IV pumps, cell phone, pagers (if used in room) and other electronics, supplies and equipment prior to removing from patient's room.

Keep essential supplies to the minimum inside the room.

Linen Management:

Line the hamper with a water-soluble red bag and keep in the patient room

Personal Protective Equipment (PPE):

Fit testing is required to determine size of N95 mask

Wear N95 mask upon entry

Remove the mask outside the room.

<u>Put ON in this order</u>	<u>Take OFF in this order</u>
1. Wash or gel hands 2. Gown 3. N95 Mask 4. Eye cover (if needed) 5. Gloves	1. Gloves 2. Wash or gel hands (even if gloves used) 3. Eye cover (if used) 4. Gown 5. Wash or gel hands 6. N95 Mask 7. Wash or gel hands

Private Room

Isolate in a negative pressure room or neutral pressure single room, keep door closed all the time

Room Cleaning:

Daily cleaning of high touch surfaces

Terminal cleaning procedure, including cubicle curtain changes

Transport:

Essential transport only.

Alert receiving department regarding patient isolation status.

Ask the patient to wear a surgical mask & to follow respiratory hygiene

Clean and disinfect transport vehicle.

Appendix (E): Airborne/ contact precautions



CONTACT/ENTERIC PRECAUTIONS

Families and visitors follow isolation instructions from information sheet
If you have questions go to nurse station

أعزائي الأقارب والزوار الرجاء اتباع إرشادات العزل المذكورة . إذا كان لديك أي استفسار الرجاء التوجه للممرضة المسؤولة



Clean hands with soap and water when entering and leaving the room. Practice the 5 moments of hand hygiene.

نظف يديك بالماء والصابون قبل وبعد دخول غرفة العزل



Wear Gown and Gloves at door

إلبس الرداء الواقي والقفازات قبل دخول غرفة العزل





Wear Mask and eye cover if there is splashing of body fluids is likely

إلبس القناع وواقي العين في حالة التعرض للرداذ والسوائل



**Use patient dedicated or disposable equipment
Clean and disinfect shared equipment**

استخدم الادوات المخصصة للمريض في غرفة العزل
نظف وعقم الادوات بين كل مريض وآخر



Royal Hospital
Infection prevention &Control

CONTACT/ ENTERIC PRECAUTIONS

Display sign outside the door, Remove sign after discontinuation of isolations

Common conditions:

Suspected/confirmed Bacterial & Viral gastroenteritis
Clostridium difficile associated diarrhea.
Cryptosporidium
Hepatitis A & E with diarrhea
Poliomyelitis
Any other conditions as recommended by Infection Control

Dishes/Utensils:

No special precautions. Kitchenware sanitized in dishwasher

Equipment and Supplies:

Use dedicated or disposable equipment when available
Clean and disinfect reusable equipment including: IV pumps, cell phone, pagers (if used in room) and other electronics, supplies and equipment prior to removing from patient's room.
Keep essential supplies to the minimum inside the room.

Linen Management:

Line the hamper with a water-soluble red bag and keep in the patient room

Personal Protective Equipment (PPE):

Don PPE upon entry
Discard before exiting the patient room in the yellow bag.

Put ON in this order

1. Wash or gel hands
2. Gown
3. Mask (if needed)
4. Eye cover (if needed)
5. Gloves

Take OFF in this order

1. Gloves
2. Wash hands with soap & water (even if gloves used)
3. Eye cover (if used)
4. Gown
5. Wash or gel hands
6. Mask
7. Wash or gel hands

Private Room

IF not available. Cohort patient that has the same organism but no other infection

Room Cleaning:

Daily cleaning of high touch surfaces
Terminal cleaning procedure, including cubicle curtain changes
Use Haz Tab for terminal cleaning in case of C.difficile diarrhea.

Transport:

Essential transport only.
Alert receiving department regarding patient isolation status.

Appendix (F): Enteric/ contact precautions



CONTACT/ENTERIC PRECAUTIONS

Families and visitors follow isolation instructions from information sheet
If you have questions go to nurse station
أعزالي الأقارب والزوار الرجاء إتباع إرشادات العزل المذكورة . إذا كان لديك أي إستفسار الرجاء التوجه للممرضة المسؤولة



Clean hands with soap and water when entering and leaving the room. Practice the 5 moments of hand hygiene.
نظف يديك بالصابون والماء قبل وبعد دخول غرفة العزل



Wear Gown and Gloves at door
إلبس الرداء الواقي والملابس قبل دخول غرفة العزل



Wear Mask and eye cover if there is splashing of body fluids is likely
إلبس القناع وواقى العين في حالة التعرض للرشاش والسوائل



**Use patient dedicated or disposable equipment
Clean and disinfect shared equipment**
استخدم الادوات المخصصة للمريض في غرفة العزل
نظف وتعقم الادوات بين كل مريض وآخر



Royal Hospital
Infection prevention & Control

CONTACT/ ENTERIC PRECAUTIONS

Display sign outside the door. Remove sign after discontinuation of isolations

Common conditions:

Suspected/confirmed Bacterial & Viral gastroenteritis
Clostridium difficile associated diarrhea.
Cryptosporidium
Hepatitis A & E with diarrhea
Polio myelitis
Any other conditions as recommended by Infection Control

Dishes/Utensils:

No special precautions. Kitchenware sanitized in dishwasher

Equipment and Supplies:

Use dedicated or disposable equipment when available
Clean and disinfect reusable equipment including: IV pumps, cell phone, pagers (if used in room) and other electronics, supplies and equipment prior to removing from patient's room.
Keep essential supplies to the minimum inside the room.

Linen Management:

Line the hamper with a water-soluble red bag and keep in the patient room

Personal Protective Equipment (PPE):

Don PPE upon entry
Discard before exiting the patient room in the yellow bag.

Put ON in this order

1. Wash or gel hands
2. Gown
3. Mask (if needed)
4. Eye cover (if needed)
5. Gloves

Take OFF in this order

1. Gloves
2. Wash hands with soap & water (even if gloves used)
3. Eye cover (if used)
4. Gown
5. Wash or gel hands
6. Mask
7. Wash or gel hands

Private Room

IF not available. Cohort patient that has the same organism but no other infection

Room Cleaning:

Daily cleaning of high touch surfaces
Terminal cleaning procedure, including cubicle curtain changes
Use Haz Tab for terminal cleaning in case of C.difficile diarrhea.

Transport:

Essential transport only.
Alert receiving department regarding patient isolation status.

Appendix G : Donning of PPE

SEQUENCE FOR PUTTING ON PERSONAL PROTECTIVE EQUIPMENT (PPE)

The type of PPE used will vary based on the level of precautions required, such as standard and contact, droplet or airborne infection isolation precautions. The procedure for putting on and removing PPE should be tailored to the specific type of PPE.

1. GOWN

- Fully cover torso from neck to knees, arms to end of wrists, and wrap around the back
- Fasten in back of neck and waist



2. MASK OR RESPIRATOR

- Secure ties or elastic bands at middle of head and neck
- Fit flexible band to nose bridge
- Fit snug to face and below chin
- Fit-check respirator



3. GOGGLES OR FACE SHIELD

- Place over face and eyes and adjust to fit



4. GLOVES

- Extend to cover wrist of isolation gown



USE SAFE WORK PRACTICES TO PROTECT YOURSELF AND LIMIT THE SPREAD OF CONTAMINATION

- Keep hands away from face
- Limit surfaces touched
- Change gloves when torn or heavily contaminated
- Perform hand hygiene



Appendix H – Donning of PPE

**HOW TO SAFELY REMOVE PERSONAL PROTECTIVE EQUIPMENT (PPE)
EXAMPLE 2**

Here is another way to safely remove PPE without contaminating your clothing, skin, or mucous membranes with potentially infectious materials. **Remove all PPE before exiting the patient room** except a respirator, if worn. Remove the respirator **after** leaving the patient room and closing the door. Remove PPE in the following sequence:

- 1. GOWN AND GLOVES**
 - Gown front and sleeves and the outside of gloves are contaminated!
 - If your hands get contaminated during gown or glove removal, immediately wash your hands or use an alcohol-based hand sanitizer
 - Grasp the gown in the front and pull away from your body so that the ties break, touching outside of gown only with gloved hands
 - While removing the gown, fold or roll the gown inside-out into a bundle
 - As you are removing the gown, peel off your gloves: at the same time, only touching the inside of the gloves and gown with your bare hands. Place the gown and gloves into a waste container
- 2. GOGGLES OR FACE SHIELD**
 - Outside of goggles or face shield are contaminated!
 - If your hands get contaminated during goggle or face shield removal, immediately wash your hands or use an alcohol-based hand sanitizer
 - Remove goggles or face shield from the back by lifting head band and without touching the front of the goggles or face shield
 - If the item is reusable, place in designated receptacle for reprocessing. Otherwise, discard in a waste container
- 3. MASK OR RESPIRATOR**
 - Front of mask/respirator is contaminated — DO NOT TOUCH!
 - If your hands get contaminated during mask/respirator removal, immediately wash your hands or use an alcohol-based hand sanitizer
 - Grasp bottom ties or elastics of the mask/respirator, then the ones at the top, and remove without touching the front
 - Discard in a waste container
- 4. WASH HANDS OR USE AN ALCOHOL-BASED HAND SANITIZER IMMEDIATELY AFTER REMOVING ALL PPE**


PERFORM HAND HYGIENE BETWEEN STEPS IF HANDS BECOME CONTAMINATED AND IMMEDIATELY AFTER REMOVING ALL PPE



Revision History			
Revision	Description of change	Reviewed by	Revision Date
1	Initial Release	Aisha Am Mamary	16/11/2020
2	Reviewed by	Dr. Omaira Al Omari	12/12/2022
3	Reviewed by	Dr. Amal Al Tai	11/06/2024
Written by		Checked by	
Aisha Al Mamary		Dr. Dr. Amal Al Tai	
		Authorized by	
		Dr. Amal Al Tai Director of IP&C	

Purpose of policy

The aim of this policy is to provide guidance for staff on the indications for isolation precautions and the risk assessment that should be performed prior to decision to isolate.

Isolation precautions should be used for patients who are:

- Either known or suspected to have an infectious disease
- Carrying a Multi-Drug Resistant Organism (MDRO)
- Particularly vulnerable to infection.

The main objectives:

- To ensure minimizing spread of infection to patients and healthcare personnel
- To ensure optimal use of isolation facilities through risk assessment
- To provide the optimal care for infected patients with minimal risk of transmission without compromising their safety

However, standard Infection Prevention control precautions must always be used for all patients regardless of their status (refer to policy on standard precautions). These include the use of gloves, aprons, masks and visors following a risk assessment to identify the risks of exposure to blood, body fluids and microorganisms.

Definition

The term “Isolation” is the use of infection prevention and control precautions aimed at controlling and preventing the spread of infection. There are two types of isolation – Source Isolation (barrier nursing) where the patient is the source of infection and Protective Isolation (reverse barrier nursing) where the patient requires protection i.e. they are immune-compromised.

Scope

This policy includes isolation precautions for both source and protective indications in the Royal hospital at both inpatients, outpatient settings and during transportation of patients. It also includes risk assessment required to assess necessity for isolation. Isolation precautions will be discussed in isolation precaution policy.

Roles & Responsibilities

All staffs working at Royal hospital are responsible for adhering to this policy and for prevention of transmission of infectious diseases by:

1. Treating all bodily fluids as potentially infectious
2. Disposing waste according to hospital policy (refer to policy on management of clinical Waste).
3. Adhering to aseptic technique when required (refer to policy on Aseptic Technique) & Aseptic Technique NUR-CF-SOP-3-Vers.1.7
4. Adhering to hand hygiene practices (refer to policy Hand Hygiene)
5. Maintaining good personal hygiene
6. Observing effective housekeeping practices
7. Adhering to standard precautions (refer to policy on Standard Precautions)

Ward and Department Managers are responsible for ensuring implementation within their area, and that all staffs working within the area adhere to the principles at all times

Bed Managers are responsible for ensuring patients are placed in accordance with this policy, and for escalating any situations where safe placement cannot be achieved. They also should coordinate with infection control practitioners to help with the risk assessment.

Administration is responsible for monitoring implementation of this policy and for ensuring action is taken when staff fails to comply with the policy.

Microbiology laboratory is responsible about detection of infectious agents and reporting results on timely manners to caring team and infection prevention and control.

The Infection Prevention & Control (IP&C)Team is responsible for:

- Providing expert advice in accordance with this policy,
- Supporting staff in its implementation
- Assisting with risk assessment where complex decisions are required.
- Ensuring this policy remains consistent with the evidence-base for safe practice, and for reviewing the policy on a regular basis.

Isolation Facilities

Source isolation is designed to prevent the spread of pathogens from an infected patient to other patients, hospital personnel and visitors.

Source isolation can be achieved by placing patients in:

- 1- **Single rooms** on general wards and multi-specialty ward, with ensuite facilities with or without anteroom.
- 2- **Negative pressure:** These are HEPA- filtered rooms used for high-risk disease/ airborne infection categories, e.g. SARS, Avian influenza, TB, viral hemorrhagic fevers, chickenpox. These rooms are located in the multi-specialty ward and in adult ICU1 and paediatric ICU.
- 3- **Positive pressure rooms** are available in the oncology center that can be used for reverse isolation.
- 4- **Cohort:**
 - When single rooms/isolation rooms are not available and where several patients with the same confirmed organism have been identified, these patients may be placed together in a bay, e.g. Clostridium difficile diarrhea or Methicillin Resistant Staphylococcus Aureus (MRSA). There is a dedicated cohort ward (FM2) to admit all patients colonized or infected with Multi resistant organisms. Please refer to Cohorting Patients for Isolation Purpose Policy (IPC-POL-2-Vers.1.0).
 - Patients tested MDRO positive two or more years ago and not admitted to any hospital after that positive result should not be admitted in cohort ward unless the repeat MDRO screening is still positive.

Criteria for Source Isolation

Patients admitted with the following symptoms must be isolated on admission:

- Known or suspected communicable infection /disease e.g. Pulmonary Tuberculosis, Chicken Pox, viral hemorrhagic fever.
- Unexplained rash if considered to be of an infectious cause
- Multi-Resistant organism's (MDROs) e.g. MRSA, Multi resistant Acinetobacter baumannii (MRAB) and Carbapenem resistant Enterobacteriaceae (CRE).
- ESBL producing coliforms must be isolated in wards that have surgical patients such as surgical wards, maternity wards, gynecology ward, nephrology, urology ward and paediatric surgical ward.
- Patients who are fitting the screening criteria for MDROs should be isolated until the

- screening results are ready (refer to screening policy).
- Diarrhea and/or vomiting until microbiologically proven negative or symptoms subside.
- Suspected viral respiratory illness such as flu, MERS-CoV etc.

If this is not possible because the patient would be at a greater risk by being isolated please contact the Infection Prevention and Control Team for further advice.

Risk Assessment

All patients identified with infectious diseases or alert organisms will be risk assessed for the need for isolation. This will take place between the Infection Prevention and Control Team and the clinical team.

Risk assessment is the assessment of the factors that influence the transmission of a pathogen and its impact. It enables staff to prioritize the use of isolation facilities.

However, the need for isolation of specific infections in side rooms does not take into account the limited resources available, which may lead to inconsistent decisions.

In order to minimize inconsistent decisions, a risk assessment must be undertaken. A discussion **must** take place between the clinical team and Infection Prevention and Control Team.

The following factors will be considered:

- The class of the pathogen and whether there is an effective treatment or vaccine e.g. prioritizing viral hemorrhagic fever over MRSA colonized patient.
- The probable route of transmission and evidence of transmission. Example: prioritize airborne infection such as TB over contact isolation due to Multi resistant organism.
- Susceptibility of the other patients near to the infected patient in the same bay i.e. do the other patients have open wounds or an invasive device. Example isolation of ESBLs in surgical wards and not in medical wards.
- Whether the organism is antibiotic resistant.
- Possible detrimental effects of isolation to the patient i.e. risk of falls, confusion or depression weighed against severity of the risk of transmission to other patients. Isolation should not compromise patient safety and care.

Risk assessment methodology is shown in **Appendix 1**.

Source Isolation Procedure:

- Place an isolation sign outside the door or a reminder sign over the bed space. The isolation sign should be self-explanatory with the type of precautions needed for each case.
- Consider what equipment and supplies are for the area and the patients care.
- Remove all non-essential furniture. The remaining furniture should be easy to clean and should not conceal or retain dirt or moisture either within or around it.
- Ensure that the hand basin has sufficient soap and paper towels for staff use.
- Ensure Alcohol hand gels are full.
- Place yellow clinical waste bag in the room on a foot-operated bin. The bag must be sealed before it is removed from the room. For patients isolated in main bays clinical waste should be placed into a small yellow waste bag and taken to the nearest clinical waste bin
- Keep the patient's personal property to a minimum. All belongings should be washable, cleanable or disposable.

- As far as is reasonably possible provide the patient with his/her own equipment i.e. commodes, sphygmomanometer etc., and all items necessary for attending to personal hygiene. Use disposable items whenever possible. Reusable equipment must be thoroughly decontaminated before being used for another patient.
- Keep dressing solutions, creams and lotions etc., to a minimum and store them within the room. These must be single patient use only
- Personal protective equipment should be kept on a cabinet placed outside the patient room to avoid contamination. Try to keep the minimum required amount. The trolley must be emptied daily and cleaned with Haztab solution.

Entering the room

- Collect all equipment needed.
- Remove any outer clothing and roll up long sleeves to the elbow.
- Put on a disposable plastic yellow apron before entering the room.
- Put on appropriate PPE as per infection control recommendation (refer to Transmission Based Precaution policy)
- Put on disposable gloves only if you are intending to deal with blood, excreta or contaminated material. Gloves do not need to be worn for routine entry into the isolation room.
- Safety glasses, visors or goggles must be available and must be worn when there is a risk of splashing of blood and body fluids.
- Wash hands with soap and water or use Alcohol hand rub before entering the room. N.B. Hand sanitizer Gel must not be used when C.diff or viral diarrhea is known or suspected
- Enter the room, shutting the door behind you.

Attending to the patient

- The patient and relatives must be informed by the care giver about the need for isolation. A full explanation of procedures and precautions must be provided. When needed, the Infection Prevention and Control Team can be contacted to discuss these issues with the patient.
- Ensure the patients are fully aware of the hospital hand hygiene policy and that they feel comfortable to ask staff and visitors to decontaminate their hands if this has not taken place.
- Meals – Whenever possible the patient's meal/water jug should be passed to another member of staff looking after the patient to take into the isolation room. If this is not possible staff serving meals/delivering water jugs should organize the delivery to deliver the meal/jug last. The meal apron can then be disposed of and hands washed on leaving the room.
- The patient must be offered the facilities to decontaminate their hands before eating
- Trays from isolation rooms should be collected last and placed directly onto the trolley. The apron must be removed after the tray is returned and hands MUST be washed. Gloves are NOT required for tray removal.
- Disposable meal trays should be provided to isolated patient.

Excreta.

- Ideally, a toilet should be kept solely for the patient's use. If neither this nor disposable items are available, a separate bedpan or urinal and commode should be left in the patient's room.
- Staff when dealing with excreta must wear gloves and aprons.
- Bedpans and urinals should be covered and taken immediately to the sluice.
- Staff must avoid touching door handles or other surfaces or items. PPE must be removed in the sluice and hands washed before returning to the isolation room where clean PPE should be worn. Commodes must be decontaminated using Haztab solution.

Accidental spills.

Any suspected contaminated fluids must be dealt with immediately according to the hospital policy.

Bathing.

If en-suite facilities are not available an infected patient must be bathed/showered last on the ward. Clean and dry the bath after the previous patient and after the infected patient.

Dressings.

- Aseptic technique must be used for changing all dressings.
- Waste materials and dirty dressings should be discarded in the appropriate yellow clinical waste bag inside the room.
- Used lotions, creams, etc., must be kept in the room and not used for other patients).
- Sterile packs must be stored safely to protect them from contamination and damage. Please refer to hospital aseptic technique policy.

Linen.

- Place infected linen in a red alginate polythene bag, which must be secured tightly before it leaves the room.
- Just outside the room, place this bag into a red linen bag, which must be secured tightly and not used for other patients. These bags should await the laundry collection in a safe area.

Waste:

Yellow clinical waste bags should be kept in the isolation room for disposal of all waste generated in the isolation room. The bag should be sealed well before leaving the room.

Leaving the room

- If wearing gloves, remove and discard them in the yellow clinical waste bag.
- Remove apron and discard it in the appropriate bag. Wash hands or use Alcohol hand rub
- Leave the room, shutting the door behind you.
- Wash hands with Soap and water or rub hands with alcohol hand rub.

Daily Cleaning of the Room

- NB. It is of paramount importance that the room or bay or bed space is cleaned daily. Housekeeping staff must be given access to the room or bay and ward staff must facilitate this process by working with housekeeping services to achieve high standards of cleanliness.
- Housekeeping staff must understand why isolation is required and should be

instructed on the correct procedure. Nursing staff must work in close collaboration with the cleaners to ensure correct procedures are followed.

- The area where isolation is being carried out must be cleaned last.
- Separate cleaning equipment must be kept for this area. The yellow color-coding system must be used when cleaning isolation rooms, bay or bed spaces. This will include mops and buckets. Refer to cleaning/terminal cleaning policies.
- Cleaners must wear PPE while cleaning and handling waste within the isolation room/bay or bed space.
- Disinfectant for daily cleaning: (disinfectant detergent) to be utilized for all isolation rooms except rooms occupied by C.difficile positive and VHF cases.
- Floor must be disinfected as above.
- Buckets should be filled in the domestic room or sluice they must not be filled from the clinical hand basin. Cleaning solutions must be changed after each room or bed space as a minimum.
- Cleaning solutions/disinfectant must be freshly diluted to the correct dilution following the manufacturer's instructions. Under no circumstances should products other than those agreed by Infection Prevention and Control and the housekeeping services be used.
- After use, the bucket must be washed and dried and returned to the cleaner room.
- Mop heads if not disposable should be removed after use with isolation patients and laundered in a hot wash after cleaning.
- Furniture and fittings should be damp dusted using disposable J cloths with the appropriate disinfectant solution.
- The toilet, shower and bathroom area must be cleaned at least once a day. Cleaning items must have red color-coding.
- Cloths must be disposed of in the clinical waste bin within the room

Transporting Infected Patients outside the Isolation Area to Other Departments:

- Patient with infectious conditions should only leave the ward for essential reasons.
- Inform the receiving department concerned about the diagnosis. Good communication is vital to prevent risks to others.
- Arrange for the patient to have the last appointment of the day whenever possible and ensure the receiving department are aware of the patient's impending arrival
- Any porters involved must be instructed carefully on any precautions required ensuring that the patient confidentiality is maintained.
- Porter staff entering the isolation room should put on appropriate PPE. The PPE should be removed once the patient is settled in the chair/bed or trolley and hands must be washed or decontaminated using alcohol gel on leaving the room.
- Gowns and gloves must NOT be worn to transfer patients through the hospital except in patients with viral hemorrhagic fever (VHF) and MERS Corona virus(MERS CoV), COVID 19 and for ICU patients if direct care needed to be delivered during transport.
- In some circumstances, for example tuberculosis, the patient should wear a surgical mask when leaving the room.
- On entering the department the patient must be taken straight to the procedures room and must not wait in the general waiting area.
- All staff attending the patient must wear appropriate PPE to move or handle the isolation patient while they are in the department.
- 5 moments of Hand hygiene should be always maintained.

- All equipment used in the room must be cleaned thoroughly after the patient has left the department. The trolley or chair should be cleaned after use. Disinfection wipes will be sufficient for this process.

Transportation of specimens of VHF cases:

- All specimens must be treated as biohazard
- Label with biohazard label
- Mark request form accordingly
- Double bag
- For confirmed cases all specimens will be handled at Containment Level 3.
- Transportation of samples between laboratories should be by enhanced category B (category 'A' packaging and category 'B' courier).

Transfer of isolated patients to other institutions:

- Inform the receiving hospital (infection control or bed manager) about the case and the isolation precautions necessary.
- Transfer of highly infectious cases such as MERS CoV and VHF cases to another hospital should be avoided unless it is necessary for medical care.
- If transfer is essential, the Infection Control Team at the receiving hospital and the ambulance staff must be advised in advance of the special circumstances of the transfer.
- Appropriate Personal Protective Equipment must be worn during transferring the patient in the ambulance

Discharging the Patient – Terminal Cleaning of the Room

- Nursing staff should inform the bed manager when the patient is due for transfer to another hospital to inform the hospital to provide the appropriate bed.
- Nursing staff must inform the disinfection team that the room will need to be terminally cleaned. Hydrogen peroxide vapor machine should be used as much as possible.
- Curtains should be changed before cleaning
- Nursing staff should strip the bed and clean all medical and nursing equipment with the appropriate disinfectant solution. This should then be removed from the room by nursing staff to allow the disinfection team staff to clean.
- Impervious surfaces, e.g. Tables and chairs, window sills etc., should be cleaned and disinfected thoroughly.
- Patients' lockers must be cleaned thoroughly including the inside.
- The floor must be disinfected and dried thoroughly. The bed should be pulled out to ensure the space behind the bed is cleaned thoroughly.
- The room can be reused as soon as it has been correctly and thoroughly cleaned. Allow room to dry thoroughly before use.
- For more details please refer to Terminal cleaning procedure (IPC-SOP-1-Vers.1.0)

Curtains

Curtains must be changed on termination of isolation and every two weeks if isolation continues longer than 2 weeks.

The contaminated curtains should be removed before the room is cleaned. The clean curtains

should be hung after the room has been cleaned.

Protective Isolation

Protective isolation is intended to prevent a more susceptible patient acquiring infection e.g. patients with lowered immune systems.

- This is best achieved in a positive pressure side room in oncology ward, although a general side room may be used with the door remaining closed.
- Staff/ visitors with infections including colds, flu like symptoms and active cold sores should not care for or visit the patient.
- The room/ furniture should be cleaned with detergent and water prior to admitting the patient.
- Staff must ensure that they decontaminate their hands and put on gown, gloves and mask before entering the room.
- Dustbins should be lined with black bag if the case is purely for protective isolation.

Monitoring Arrangements

Regular Monitoring

The policy will be monitored using isolation audit tool. Results will be reported to the Infection Control Committee who will review compliance and if necessary nominate a manager to develop an action plan to achieve the standards / process set out in the document. This action plan will also be monitored via IP&C Committee.

Appendix 1

Alert organism risk assessment for the use of isolation rooms

All patients suspected or known to be colonized or infected with an infectious disease or condition must be isolated, in line with Isolation Policy. However, due to limited isolation facilities it is recognized that, at times, single rooms will need to be prioritized. The following information is intended to assist Senior Nursing staff, Bed Management teams and Ward Managers in this process. For the further list of infectious diseases and the isolation precaution needed please refer to GCC infection control manual in the Royal portal pages 45-58.

Code	Action
3	High priority for a single room or isolation room with negative pressure. Inform Infection Prevention and Control Team(ICT) if single room or isolation room not available.
2	Single room required. Assess patients currently in side rooms. If single room not available, contact Infection Prevention and Control to discuss further. Where side room n/a, nurse in main bay, providing other patients in the bay are not deemed vulnerable i.e. with open wounds or invasive devices, immune-compromised etc. Move to single room ASAP.
1	Low risk. Single room or cohort bay required, but may be nursed in bay providing other patients in the bay are not deemed vulnerable i.e. with open wounds or invasive devices, immune-compromised etc
0	Does not require Isolation
*	May require contact tracing

Risk assessment for isolation of Multi-drug resistant organisms and some selected common infections.

Alert organism/assessment	Precautions	Level	Comments
Multi Drug Resistant Gram negative MDRAB, CRE (MDRO) Colonized in screening body sites i.e. nose, invasive devices, i.e. PVC Positive Sputum Oozing wound Tracheostomy	Contact	2*	Priority to go to isolation ward All Gram negative MDROs must be cohorted in the isolation ward if fitting the criteria(refer to cohort policy)
Other Multi resistant organisms i.e. ESBL Catheterized Oozing wound	Contact	2*	Isolate only those who are admitted in surgical wards or admitted near surgical patients Don't isolate cases in medical wards
MRSA Skin colonization	Contact	1*	
MRSA Positive Wound pouring pus Sputum with cough or tracheostomy	Contact	2*	Prioritize MRSA to go to single rooms or if MRSA cohort is available can be cohorted
High risk patients awaiting body screening for MDROs As per screening policy	Contact	2*	Apply precautions until all results are available
Diarrheal/ vomiting illness suspected to be infectious C.difficile, Campylobacter Norovirus etc.	Contact enteric precautions	2	Vomiting should precedes diarrhea for isolation if single rooms are limited
Chickenpox Rash developed within previous 10 days	Airborne and contact	3*	Only staff with a history of Chicken Pox should have contact with patient

			Isolate until all lesions are crusted
Chickenpox Rash dry, longer than 10 days old	Standard	0	
Shingles Rash in an exposed area Such as face	Airborne and contact	3*	Isolate until lesions are crusted
Shingles Rash is disseminated to 3 more dermatomes	Airborne and contact	3*	Isolate until lesions are crusted
Shingles Localized in immune-compromised	Airborne and contact	3*	Until disseminated zoster is ruled out, then can revert to standard precautions
Shingles Localized and covered in immune-competent	Standard	0	Cover lesions, wear gloves if contact with lesions
Confirmed pulmonary TB	Airborne	3*	Until 14 days of treatment With 3 negative AFB smears and clinical /radiological improvement
Suspected Pulmonary TB	Airborne	3*	Isolate until three AFB sputum samples are negative on microscopy If low risk 1 negative TB PCR(ultrasensitive) If high risk 2 negative PCR or send more samples for microscopy.
Suspected or confirmed MDR Pulmonary TB	Airborne	3*	Requires negative pressure ventilation in multi-specialty ward (priority), consult infection control for de-isolation
Extra-pulmonary TB Aerosolizing procedures (surgery) or draining wounds Inter-costal drain	Airborne	3*	
Extra-pulmonary TB Without risk of aerosol	Standard	0	
Scabies	Contact precautions	2*	
Group A Strep	Droplet/contact	2	Until 48 hours of appropriate antibiotics

Head Lice	Contact precautions	2	Clothing and linen used by the infected patient from 3 days prior and 24 hours after treatment must be placed in a hot water soluble bag labeled and sent to laundry.
Suspected meningococcal Meningitis	Droplet	3*	Until 24 hours of antibiotics. Inform Infection Prevention and Control if indirect contact with respiratory secretions during resuscitation
Influenza / other viral respiratory infections	Droplet	2*	Until 4 days after start of Flu treatment or until clear of symptoms (for critical care or immune-compromised patient discuss with infection control)
Measles	Airborne	3*	Isolate until 4 days after appearance of the rash For duration of illness in immune-compromised Assign immune HCWs to take care of measles cases
Mumps	Droplet	2*	Until 9 days from appearance of parotitis
Rubella	Droplet	2*	Until 7 days after onset of rash
MERS corona	Airborne	3*	Isolate for the duration of illness
Viral hemorrhagic fever	Airborne/contact precautions	3*	Inform infection Prevention and Control or Microbiologist on call immediately if diagnosis is suspected

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
GCC Infection Prevention & Control Manual	GCC center for infection control	2013	58-68
Guideline for Isolation Precautions: Preventing Transmission of Infectious Agents in Healthcare Settings	CDC	2007	18, 71-72
Heart of England NHS Foundation Trust Isolation Policy	Trust's Infection Prevention and Control Team	2010	1-39

Review / Revision History

Revision History			
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